

Yelman Khan

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Research Focus: Empirical AI security and safety, with an emphasis on **representation learning** for device and vulnerability intelligence, **LLM-based systems** evaluated on real-world cybersecurity data, and improving the realism and reliability of **security benchmarks**.

EDUCATION

University of Massachusetts Amherst

Aug. 2020 – May 2022

MS in Computer Science; GPA: 3.74

Courses: Advanced NLP, Machine Learning, Advanced Algorithms, Database Design & Implementation

Rochester Institute of Technology

Aug. 2015 – Dec. 2019

BS in Computer Engineering; Minor in Mathematics; GPA: 3.61

Courses: Graph Theory, Data Mining, Data & Communications Networks, Computer Architecture & Organization

EXPERIENCE

Palo Alto Networks

Jun. 2022 – Present

IoT Machine Learning Software Engineer

Santa Clara, CA

- **Security Research & Threat Analysis:** One of three main contributors to the [Device Security Threat Report 2025](#), analyzing real-world **device telemetry** and **vulnerability intelligence** to characterize emerging IoT threats.
- **LLM Systems & Evaluation:** Built and evaluated a customer-facing security Copilot using **large language models** over structured device and vulnerability data, with emphasis on **evaluation**, **failure analysis**, and retrieval-based robustness.
- **Behavior Modeling & Policy Recommendation:** Developed a **representation learning** pipeline using **embeddings**, **clustering**, and **anomaly detection** to identify common device behaviors; added an **LLM-based confidence scoring** layer to reduce unsafe or ambiguous policy recommendations.
- **Namesync (Identity Systems):** Built an embedding-based identity alignment system using **contrastive learning** and **vector similarity search** to match internal device entities with external references across **MDS2 medical device data** and vulnerability datasets, with **LLM-assisted evaluation** of ambiguous matches.

Retail Business Services

Jun. 2019 – Jun. 2020

Software Engineer

Quincy, MA

- Built a pipeline to convert retail CAD floor plans into interactive VR store environments (Unity/C#).
- Developed an ML model to predict inventory reject rates with **0.9 precision**.

PATENTS & PUBLICATIONS

- **IoT policy recommendation LLM embeddings for global behavior learning** — [US12278731B1](#) (Granted)
- **IoT security knowledge-based chatbot system** — [US12475115B2](#) (Granted)
- **Maruna Bot: An Extensible Retrieval-Focused Framework for Task-Oriented Dialogues** — Alexa Prize TaskBot Challenge Proceedings, 2022
- **Combinatorial Scientific Discovery: Finding New Concept Combinations Beyond Link Prediction** — ICLR 2022 Workshop on Deep Learning on Graphs for NLP

TECHNICAL SKILLS

ML & Research: PyTorch, **contrastive learning**, embeddings, clustering, anomaly detection, evaluation design

LLM Systems: retrieval-augmented generation, in-context learning, **LLM-based evaluation**

Data & Infrastructure: SQL, vector search, large-scale pipelines, AWS, GCP, Docker

Languages: Python, SQL, C++